



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

PROJECT TITLE

IPL Auction Team Selection Using Optimization Algorithms

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the course

CSA-0603 Design And Analysis Of Algorithms

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN CSE/ARTIFICIAL INTELLEGEENCE & DATA
SCIENCE**

Submitted by

CH.V.Rama Yogi(192424319)

D.Hussainaiah(192425300)

K.Subramanyam(192411253)

A.Velmurugan(192521074)

Under the Supervision of

Dr. P. Solainayagi

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “IPL Auction Team Selection Using Optimization Algorithms” has been carried out by CH.V.Ramayogi (192424319) D.Hussainaiah(192425300) , K.Subramanyam(192411253), A.Velmurugan(192521074) under the supervision of Dr. P. Solainayagi and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech CSE/AIDS program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.F.Mary Harlin Fernandez

Professor

Department Of CSE

SIMATS Engineering,

SIMATS, Chennai – 602105.

COURSE FACULTY

Dr. P. Solainayagi

Professor

Department Of CSE

SIMATS Engineering,

SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

I, Ch.Venkata Ramayogi, D.Hussainaiah, K.Subramanyam , A.Velmurugan of the Department of CSE/AIDS, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “IPL Auction Team Selection Using Optimization Algorithms” is the result of my own bonafide efforts. The work presented herein is original and has been carried out in accordance with engineering ethics and academic integrity. All sources, references, data, software resources, datasets, and AI-assisted resources used in the project have been appropriately acknowledged. The data, results, analysis, and findings have not been fabricated, falsified, or manipulated.

Place: Chennai

Date: September 2026

Signature of the Students with Names

Ch.Venkata Ramayogi

D.Hussainaiah,

K.Subramanyam ,

A.Velmurugan

INDIVIDUAL CONTRIBUTION STATEMENT

Student / Reg. No.	Responsibilities	Design & Development	Testing & Analysis	Report Contribution	%
Ch. Venkata Ramayogi / 192424319	System architecture, frontend and integration	UI, architecture, integration	System testing	Chapters, diagrams	40%

ABSTRACT

IPL auctions require teams to select players while balancing a limited purse, player performance, squad composition and auction constraints. This project proposes a web-based IPL Auction Team Selection system using Design and Analysis of Algorithms (DAA) optimization techniques. The system manages players, teams, auction bidding, budgets and squad composition through a responsive interface. The core optimization engine applies the 0/1 Knapsack algorithm to maximize overall team performance within a given budget, while a Greedy algorithm provides a fast heuristic for player selection. Sorting and Priority Queue techniques support player ranking and live bidding operations. The application provides team recommendations, Playing XI selection, auction history, budget analytics and performance dashboards. Testing is performed using controlled IPL-style player datasets and simulated auction scenarios. The expected outcome is an efficient and transparent decision-support system that reduces manual effort and demonstrates practical optimization algorithms in a real-world sports management problem.

Keywords: IPL Auction, Optimization Algorithms, 0/1 Knapsack, Greedy Algorithm, Dynamic Programming, DAA, Team Selection, Budget Optimization, Web Application

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Meaning
DAA	Design and Analysis of Algorithms	Algorithm design and complexity analysis
IPL	Indian Premier League	Professional cricket league
DP	Dynamic Programming	Optimization technique
API	Application Programming Interface	Software communication interface
UI	User Interface	Application interaction layer
DB	Database	Persistent relational storage
XI	Playing XI	Selected eleven players

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

An IPL auction is a constrained decision-making process in which teams compete for players while working within a fixed purse and squad rules. Selecting players manually becomes difficult when many candidates have different prices, roles and performance levels. An algorithmic system can model player selection as an optimization problem and recommend a high-value squad while respecting budget and composition constraints.

1.2 Need for the Project

- Manual player selection can take more time when many players are available.
- Teams must balance player performance, cost and role requirements simultaneously.
- A large player pool creates many possible squad combinations that are difficult to compare manually.
- Team owners need quick access to player statistics, price and auction status.
- Optimization provides systematic and repeatable team-selection decisions.
- The project combines a responsive web interface, Flask backend, MySQL database and DAA optimization engine.

1.3 Problem Statement

IPL auction team selection is difficult because many players have different prices, roles and performance values, while teams must stay within a fixed purse and satisfy squad constraints. The engineering problem is to design and implement a system that manages auction activity and uses optimization algorithms to select a high-performance, balanced squad without exceeding the available budget.

1.4 Project Aim

To design and implement a reliable web-based IPL auction team-selection system that uses optimization algorithms to recommend a high-performance squad and Playing XI within budget and configured squad constraints.

1.5 Project Objectives

1. To design a web-based IPL auction management system with a structured database and clear user roles.

2. To implement player search, filtering, sorting and performance-based ranking.
3. To implement secure team, player, budget and auction data management.
4. To validate bids, prevent invalid purchases and maintain consistent squad and purse information.
5. To generate optimized squad recommendations, Playing XI suggestions, auction history and analytics reports.
6. To test optimization and auction operations using controlled IPL-style player datasets and simulated auction scenarios.

1.6 Scope

- Includes player ID, name, role, nationality, base price, current bid and performance statistics.
- Includes Player Management, Auction Management, Team Optimization and Dashboard & Analytics.
- Uses MySQL for normalized persistent storage of users, teams, players, auctions, bids, squads and analytics.
- Large-scale commercial deployment, continuously changing real-world IPL feeds and advanced predictive analytics are outside the academic prototype scope.
- Assumes valid player data, authenticated users and configured auction rules and constraints.

1.7 Expected Engineering Outcome

The intended outcome is a working prototype that demonstrates DAA optimization through a real-time auction interface, valid budget-aware squad recommendations, Playing XI selection, team analytics and functional validation.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was organized around recent work in IPL player valuation, sports analytics, squad optimization, multi-objective optimization and auction strategy. These studies provide methods for measuring player value and constructing balanced squads.

2.2 Review of Existing Work

Author & Year	Title / Study	Methodology	Advantages	Disadvantages
Malhotra (2022)	Predicting IPL Auction Prices and Player Value	Player valuation and data analysis	Supports objective player valuation	Does not directly optimize a complete squad
Verma et al. (2022)	A Balanced Squad for Indian Premier League using Modified NSGA-II	Multi-objective evolutionary optimization	Balances multiple squad objectives	Higher computational complexity
Dey et al. (2024)	Optimising IPL Squad Composition	Mathematical optimization framework	Models budget and squad constraints	Depends on chosen performance metrics
Dave & Patel (2025)	Multi-Objective Optimization for IPL Team Selection	Real-binary optimization	Supports balanced team construction	Sensitive to selected indicators
Joglekar et al. (2025)	Clustered Auction Strategies and Performance Outcomes	Clustering and sports analytics	Provides auction strategy insights	Focuses more on analysis than automated optimization

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Manual selection	Flexible human judgement	Slow and difficult for large player pools	Motivates algorithmic assistance
Spreadsheet selection	Easy filtering and comparison	Limited optimization and real-time interaction	Useful baseline
Rule-based selection	Fast and simple	May miss better player combinations	Baseline heuristic
Optimization-based system	Systematic constrained selection	Requires objective and constraints	Directly supports project

2.4 Research / Engineering Gap

The project addresses a practical gap by integrating auction interaction, budget management and algorithmic squad optimization in one web application. Existing approaches often focus on valuation or analysis separately, while this project connects player value, auction decisions and constrained team selection.

2.5 Proposed Contribution

The proposed system connects player statistics, auction bidding, budget tracking and DAA optimization in one platform. The optimization objective and constraints are made visible so that recommendations can be explained, tested and compared.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Admin	Manage players, teams, auction settings and reports.
Team Owner	View players, participate in bidding, manage purse and squad.
Analyst	View player statistics, team comparison and analytics.
Project Evaluator	Verify DAA algorithms, complexity, constraints and testing evidence.
System	Maintain consistent budget, auction and squad data.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Player management	Functional	Add/update/delete player records
Auction bidding	Functional	Validate bids and update purse
Optimization	Functional	Return a feasible high-value squad
Budget control	Functional	Never exceed available purse
Search and sorting	Functional	Return relevant ranked player results
Usability	Non-Functional	Clear navigation and feedback
Security	Non-Functional	Authenticated role-based access

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	Application
ISO/IEC 25010	Software quality characteristics	Usability, reliability, maintainability and performance
ISO/IEC/IEEE 29148	Requirements engineering practices	Functional and non-functional requirements

REST	API design principles	Communication between frontend and Flask backend
Academic integrity	Acknowledge sources and tools	References, dataset and AI-assisted resources disclosed

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Team Purse	Configurable auction budget	₹	High
Player Cost	Base/current bid price	₹ Cr	High
Performance Score	Normalized player value	Score	High
Squad Size	Configured maximum	Players	High
Player Role	Batter/Bowler/All-Rounder/Wicketkeeper	Category	High
Nationality	Indian / Overseas	Category	Medium
Optimization	Maximize score within constraints	Objective	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual notebook/contact list. Simple and inexpensive, but searching and updating become difficult as records grow.

Alternative 2: Database-backed application. Strong query and concurrency features, but more infrastructure is required.

Alternative 3: Proposed file-based C++ system. Low infrastructure, persistent storage, and direct demonstration of OOP and file handling.

Criterion	Weight	Manual	Rule-Based	Proposed
Selection quality	25%	Medium	Medium	High
Budget handling	20%	Low	High	High
Speed	15%	Medium	High	High
Constraint handling	20%	Low	Medium	High
DAA relevance	20%	Low	Medium	High
Overall suitability	100%	Medium-Low	Medium	High

3.5 Selected Design & Detailed Design

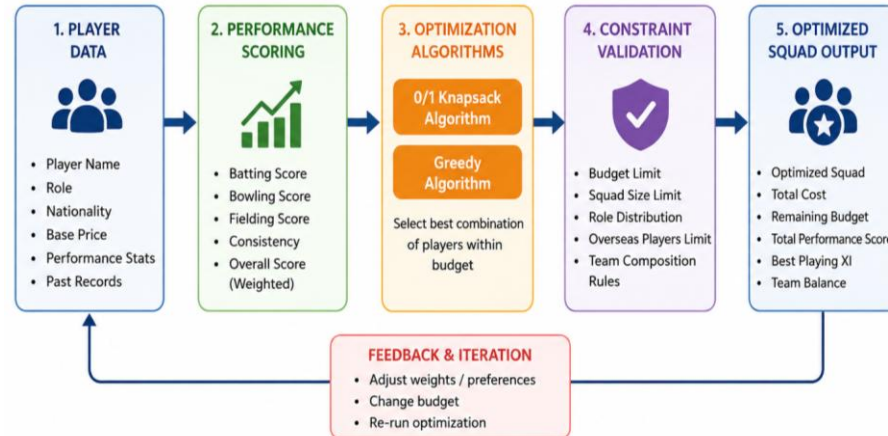


Figure 3.1 – Team Selection & Optimization Architecture

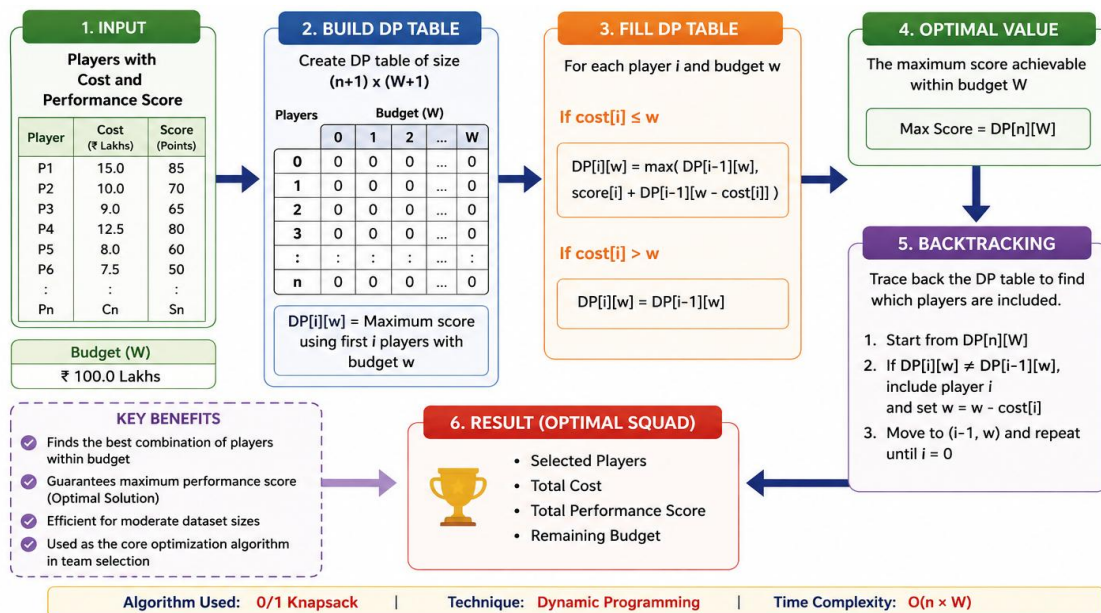


Figure 3.2 – 0/1 Knapsack Optimization Flow

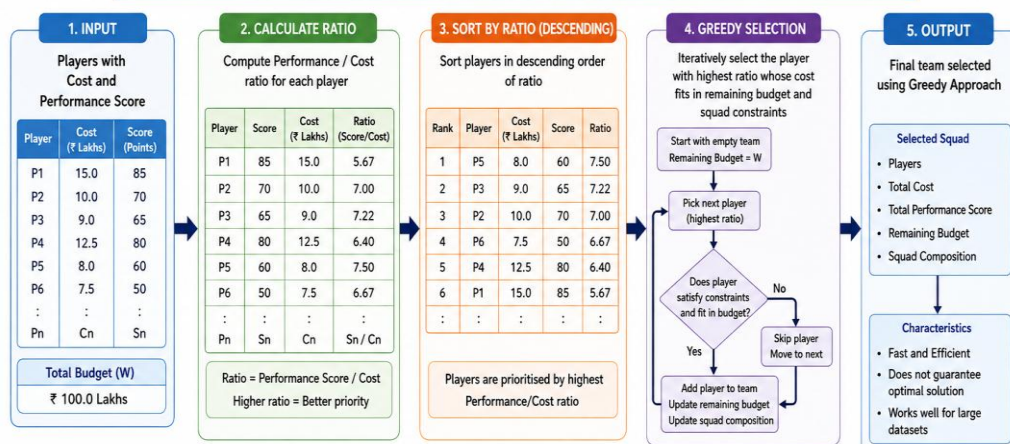


Figure 3.3 – Greedy-Based Player Selection Flow

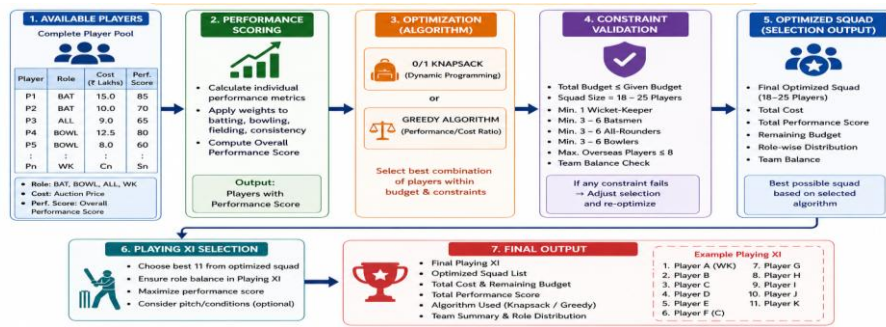


Figure 3.4 – Optimized Team Selection Process

The web-based layered design is selected because it supports real-time interaction while making the DAA concepts visible. The architecture separates the interface, backend services, auction processing, optimization engine, database and analytics.

Layer	Main Responsibility
User / Web Interface	Authentication, dashboards, auction screens and reports.
Flask API Layer	Routes requests, validates input and manages sessions.
Auction Service	Processes bids, purse updates and auction state.
Optimization Engine	Runs 0/1 Knapsack and Greedy team-selection algorithms.
Constraint Validation	Checks budget, squad size, roles and configured rules.
MySQL Database	Stores users, players, teams, bids, squads and analytics.
Analytics Layer	Generates rankings, budget and performance insights.

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
0/1 Knapsack	Exhaustive search	Exact optimization for defined budget model	Depends on budget discretization	Selected
Greedy	DP only	Very fast heuristic	Not always globally optimal	Use for comparison
MySQL	Flat files	Structured queries and persistence	Requires DB setup	Selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence	Impact
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Sustainability	Use efficient software resources	Software-only prototype	Low infrastructure requirement
Ethics	Use responsible player data	Acknowledged/public or synthetic data	Reduces data concerns
Security	Protect users and auction data	Hashing, sessions, role-based access	Improved protection
Data Integrity	Prevent invalid transactions	Server-side validation and transactions	Consistent results

Risk Register

Risk	Likelihood	Severity	Level	Mitigation	Residual
Invalid bid	Medium	High	High	Server-side bid and budget validation	Low
Budget inconsistency	Medium	High	High	Atomic database updates	Low
Incorrect optimization	Medium	High	High	Known test cases and constraint checks	Low-Medium
Unauthorized access	Low-Medium	High	High	Authentication and role authorization	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an incremental development approach. Requirements and auction constraints are defined first, followed by database design and interface implementation. Auction operations are then integrated with the optimization engine, followed by algorithm testing, system testing and analytics validation.

Resource	Purpose
HTML/CSS/JavaScript	Responsive user interface
Bootstrap 5	UI components and responsive layout
Python Flask	Backend and REST APIs
MySQL	Persistent relational storage
Chart.js	Analytics visualization
VS Code	Development and debugging

4.2 Software Implementation & Modern Tools Used

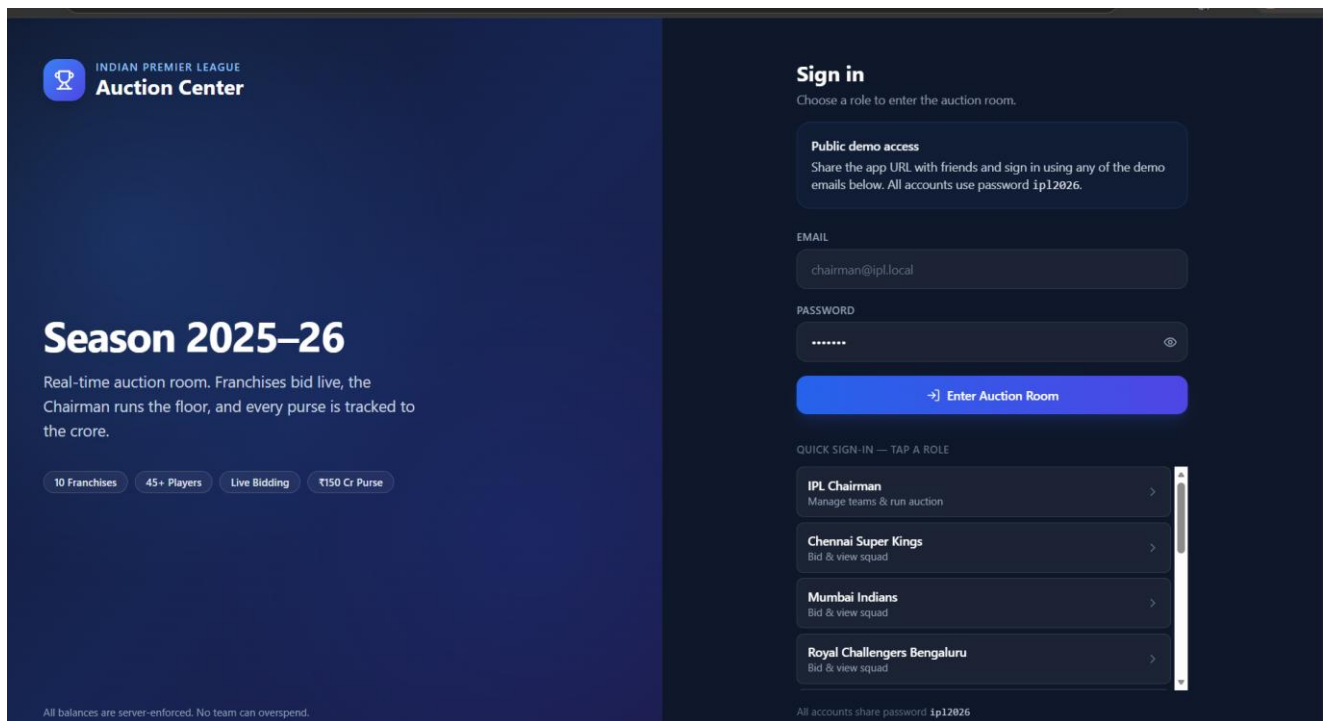


Figure 4.1 – Web Application Dashboard

Tool / Software	Purpose	Stage
Python Flask	Backend and REST API	Implementation
MySQL	Database management	Design/Implementation
Bootstrap 5	Responsive UI	Frontend
Chart.js	Analytics charts	Frontend/Analytics
Optimization Algorithms	Team selection	DAA Implementation
VS Code	Coding and debugging	All stages

4.2.1 Data Model and File Format

Each player record contains PlayerID, Name, Role, Nationality, BasePrice and performance indicators such as runs, wickets, strike rate and economy. These values are used to derive a performance/value score for optimization.

Example: P001 | Player Name | All-Rounder | Indian | 8.00 Cr | Performance Score 88 | Runs 420 | Wickets 12

4.2.2 Module Implementation

Module	Input	Processing	Output
Player Management	Player details and statistics	Store, validate, search and rank players	Player information
Auction Management	Player, bid and team data	Validate bids and update purse	Auction result and bid history
Team Optimization	Players, costs, scores, constraints	Run Knapsack and Greedy algorithms	Optimized squad and Playing XI

Module 3: Team Selection & Optimization Module

Description

This module is the main DAA component of the project. It selects the best possible IPL team from the available players based on player performance, player cost, team budget, and squad requirements. The module uses optimization algorithms to maximize team performance while satisfying the configured constraints.

Features

- **Player Selection:** Selects suitable players based on performance and price.
- **Budget Optimization:** Ensures the selected squad stays within the available purse.
- **0/1 Knapsack Algorithm:** Finds the best combination of players for maximum performance within the budget.
- **Greedy Algorithm:** Provides a fast player-selection solution based on the performance-to-cost ratio.
- **Squad Balance:** Maintains the required balance between batsmen, bowlers, all-rounders, and wicketkeepers.
- **Constraint Validation:** Checks squad size, budget, and overseas-player limits.
- **Playing XI Recommendation:** Recommends the best Playing XI from the selected squad.
- **Team Performance Score:** Calculates the overall strength of the selected team.
- **Algorithm Comparison:** Compares Greedy and 0/1 Knapsack results based on score, cost, and execution time.
- **Optimization Report:** Displays selected players, total cost, remaining budget, and performance score.

Input

The module takes the following inputs:

1. Available team budget
2. Player cost
3. Player performance score
4. Player role
5. Player nationality
6. Squad requirements
7. Configured auction constraints

Process

The team selection and optimization process follows these steps:

Player Data → Calculate Performance Score → Apply Optimization Algorithm → Check Budget & Squad Constraints → Select Best Player Combination → Calculate Team Performance → Recommend Best Playing XI

Output

The module produces the following outputs:

1. Optimized Squad
2. Total Squad Cost
3. Remaining Purse
4. Overall Performance Score
5. Squad Composition
6. Best Playing XI
7. Optimization Algorithm Used
8. Algorithm Execution Time
9. Team Recommendation Report

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Login	Valid and invalid credentials	Correct role-based access
Player search	Search by name / role	Relevant players displayed
Bid validation	Bid below/above allowed value	Invalid bid rejected
Budget control	Attempt purchase beyond purse	Transaction rejected
0/1 Knapsack	Known small test dataset	Expected optimum returned
Greedy	Known ranking dataset	Expected heuristic order
Squad constraints	Invalid composition	Constraint violation rejected

Specification	Required Value	Achieved Value	Status
Budget constraint	Never exceed purse	To be measured from final build	Pending evidence
Optimization correctness	Known test optimum	To be measured from final build	Pending evidence
Bid validation	Invalid bids rejected	To be measured from final build	Pending evidence
Authentication	Unauthorized users blocked	To be measured from final build	Pending evidence
Analytics	Correct auction totals	To be measured from final build	Pending evidence

*Update these statuses with actual observed evidence from the final application before submission.

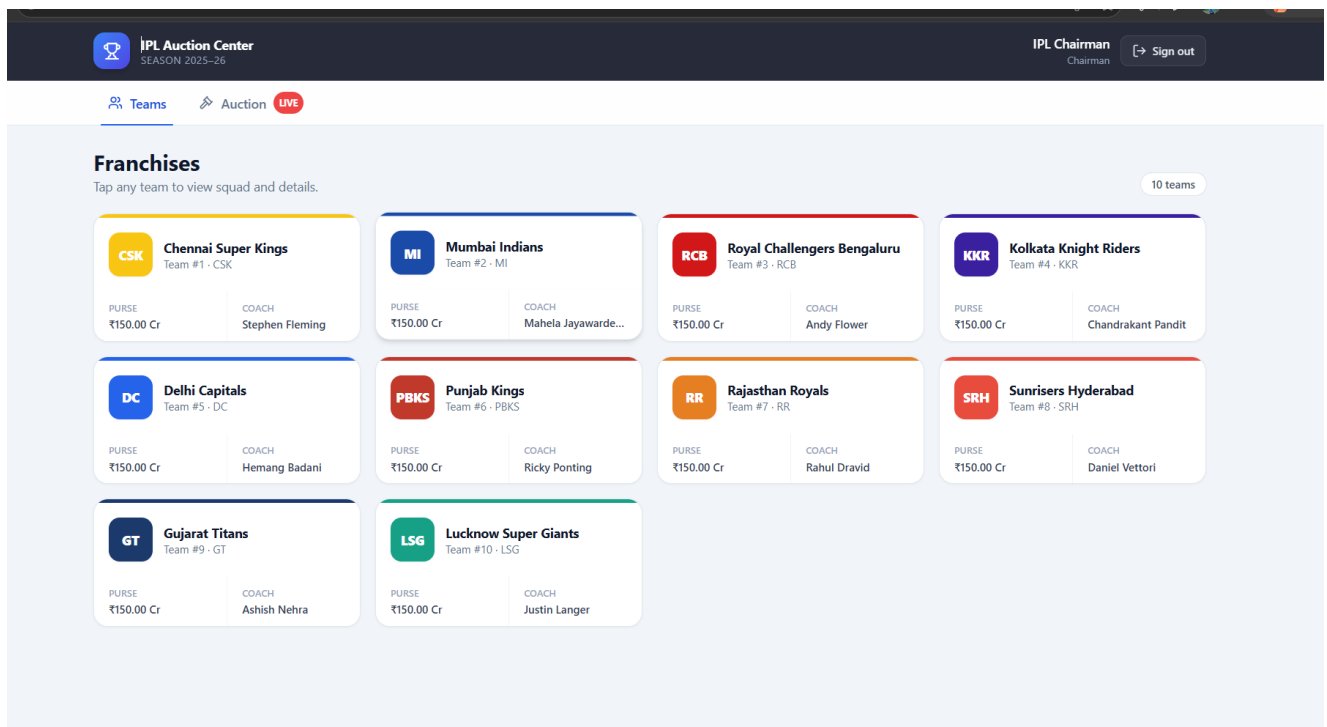


Figure 4.2 – Teams Dash board

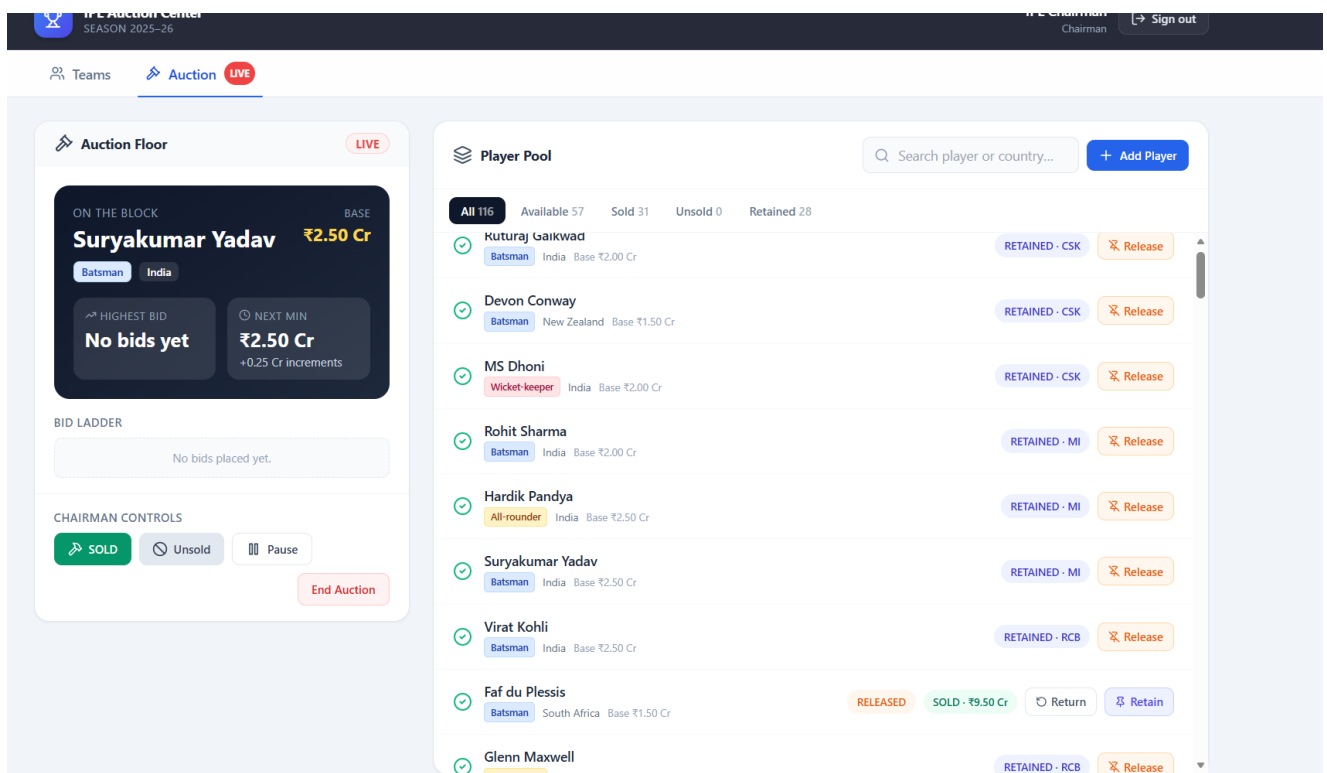


Figure 4.3 – Live Auction

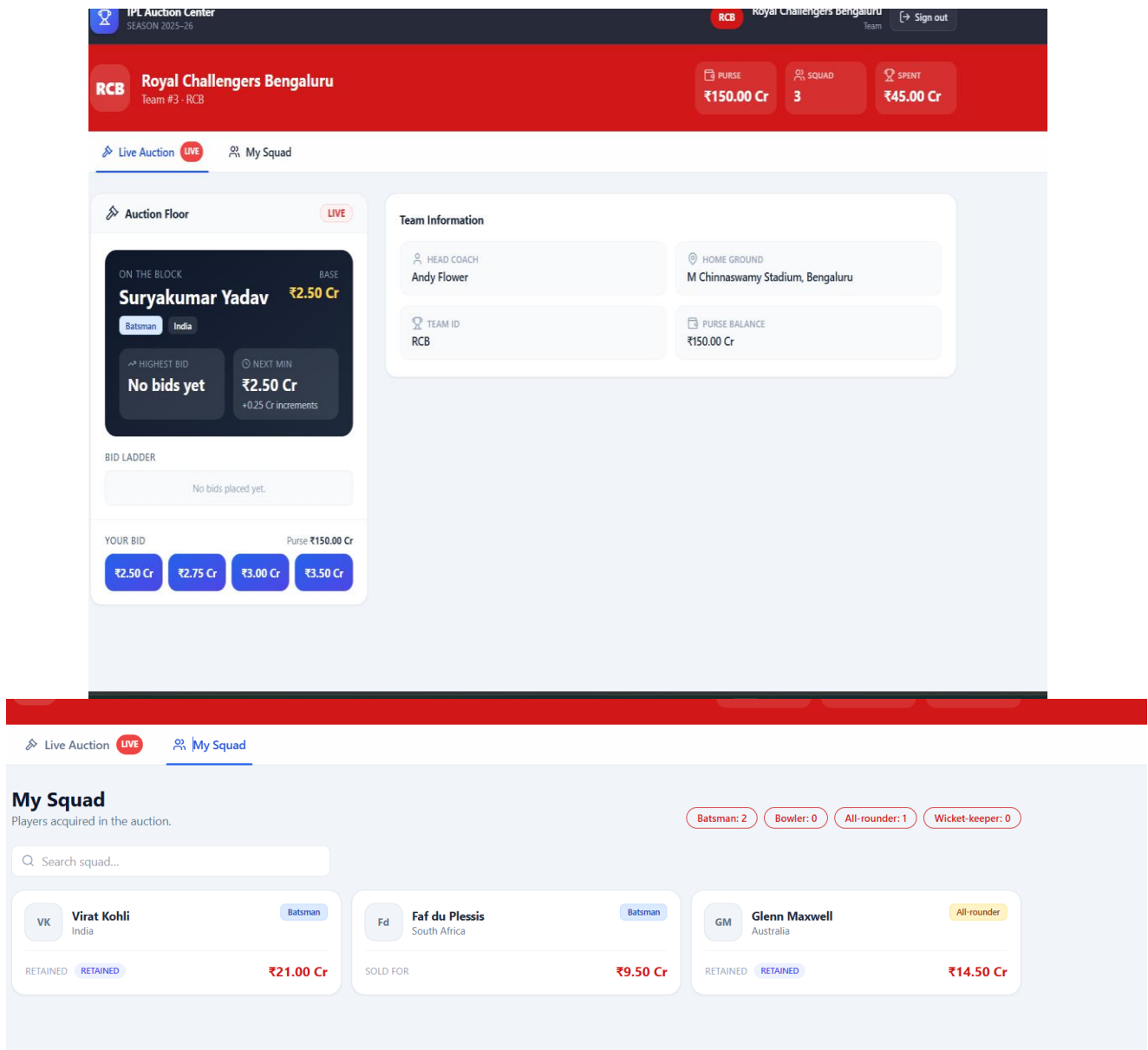


Figure 4.4 – Team Players

4.4 Results

The prototype demonstrates player management, live bidding, budget tracking, team optimization, Playing XI recommendation and analytics. Final execution times, objective values, squad cost and screenshots should be inserted from the running build.

4.5 Data Analysis & Engineering Judgment

The optimization layer is separated from the interface so algorithm correctness and performance can be evaluated independently. The 0/1 Knapsack method provides an exact solution for the defined budget-

indexed model, while the Greedy method provides a fast heuristic. For larger multi-objective formulations, more advanced optimization techniques may be considered.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Implement optimization	Knapsack + Greedy engine	Implemented / validate final build
Budget-aware team selection	Constraint validation	Implemented / validate final build
Auction management	Bid and purse services	Implemented / validate final build
Analytics	Dashboard and charts	Implemented / validate final build
Testing	Algorithm and system test cases	Planned / validate final build

4.7 Strengths & Limitations

Strengths:

- Simple file-based architecture.
- Clear three-module separation.
- Demonstrates C++ OOP and File Handling.
- No database server required.
- Synthetic data supports privacy in testing.

Limitations:

- Sequential search is less scalable than indexed queries.
- Concurrent multi-user editing is outside scope.
- Advanced authentication is limited.
- Text files provide fewer integrity guarantees than a DBMS.

4.8 Project Planning, Roles & Budget

Milestone	Activity	Status
1	Requirements and problem definition	Completed
2	Literature and architecture	Completed

3	Database and UI implementation	In progress
4	Auction engine integration	In progress
5	Optimization implementation	In progress
6	Testing, report and repository	Planned

Student	Assigned Responsibility	Actual Contribution
Ch. Venkata Ramayogi	Architecture, frontend and integration	To be updated

Item	Estimated Cost	Actual Cost
Existing laptop	Existing resource	Existing resource
Software / IDE	₹0	₹0
MySQL / Flask / Bootstrap	₹0	₹0
GitHub / version control	₹0 basic use	₹0
Dataset preparation	₹0 / academic data	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The IPL Auction Team Selection Using Optimization Algorithms system provides a structured decision-support solution for constrained team selection. It combines player management, auction bidding, budget tracking and analytics with DAA optimization techniques. The 0/1 Knapsack algorithm can identify a high-value combination within a defined budget, while the Greedy algorithm provides a fast heuristic for comparison. Sorting supports player ranking and Priority Queue operations support auction bidding. Overall, the project demonstrates how algorithm design and analysis can be applied to a practical sports-management problem.

5.2 Future Work

- Replace text-file storage with a database for larger datasets.
- Add authentication and role-based access control.
- Add sorting, filtering and duplicate detection.
- Provide automatic backup and restore.
- Add CSV import/export.
- Add cloud deployment and multi-user synchronization.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- C++ classes, objects, functions and file streams.
- File-based CRUD operations and record rewriting.
- Interface and application integration.
- Validation, testing and backup concepts.

Engineering & Professional Skills Developed:

- Problem formulation and modular design.
- Debugging and functional testing.
- Git/GitHub version control.

- Technical documentation and presentation.

Individual Reflection: The most difficult engineering problem was maintaining consistent records when an existing contact had to be updated or deleted. This was addressed by treating the text file as the persistent source and rebuilding the maintained record set after modification. A key decision was to keep storage file-based because it directly demonstrates the project requirements and avoids unnecessary database infrastructure. If redesigned for production, database storage, authentication, transactions and stronger validation would be added.

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- [10] Oracle, MySQL Documentation.
- [11] Chart.js Documentation.
- [12] MDN Web Docs, HTML, CSS and JavaScript Documentation.

APPENDICES

Appendix A – Detailed Calculations

No hardware sizing calculations are required for this software-based prototype. Record actual optimization execution time, memory usage where available, dataset size and final test results in the submission.

Appendix B – Engineering Drawings / Schematics

Step	System Flow
1	User opens the application.
2	User authenticates and selects the appropriate role.
3	Player and team data are loaded.
4	Auction starts and bids are validated.
5	Winning purchases update the team purse and squad.
6	Optimization engine evaluates candidate player selections.

Appendix C – Source Code / Code Repository Information

GitHub repository: Add the approved IPL auction project repository URL before final submission.

Complete source code may be maintained in the approved repository; include only essential excerpts in the report.

Appendix D – Dataset Information

The project dataset should contain IPL-style player records including player ID, name, role, nationality, base price and performance indicators. Use a properly acknowledged public dataset or synthetic records for testing.

Appendix E – Experimental / Raw Data

Attach the final CSV/TXT dataset and selected test logs or screenshots here if required.

Appendix F – Ethics / Institutional Approval

No human-subject experiment is proposed. Player data should be sourced responsibly and acknowledged appropriately; synthetic records may be used for algorithm testing.

Appendix G – Project Meeting / Progress Records

Date	Activity	Outcome
21-07-2026	Requirement discussion	Problem and scope finalized.
30-07-2026	Architecture review	Four functional areas and optimization approach selected.
20-08-2026	Implementation review	Core application development reviewed.
25-08-2026	Testing review	Algorithm and system test plan prepared.
07-09-2026	Final review	Report and repository prepared.

Appendix J – Individual Contribution Evidence

Ch. Venkata Ramayogi (192424319) – primary contribution included architecture, C++ OOP implementation, Search & Retrieval, Update & Deletion, Report Generation, testing, documentation and repository preparation.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location
Complex engineering problem formulation	SO1	Chapter 1, Sections 1.2–1.3
Engineering design under constraints	SO2	Chapter 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Chapter 3.7 / Appendix F
Teamwork and leadership	SO5	Chapter 4.8 / Contribution Statement
Experimentation and engineering judgement	SO6	Chapter 4.3–4.6
Independent knowledge acquisition	SO7	Chapter 5.3
Standards	SO2	Chapter 3.2
Sustainability / societal impact	SO2/SO4	Chapter 3.7

Risk assessment	SO2/SO4	Chapter 3.7
Security considerations	Relevant SO	Chapter 3.7
Integrated/system-level engineering	SO1/SO2	Chapter 3.5
Project planning	SO5	Chapter 4.8
Individual contribution	SO1–SO7	Contribution + Chapter 4.8